

11

Perspectives

Perspectives are used within business analysis work to provide focus to tasks and techniques specific to the context of the initiative. Most initiatives are likely to engage one or more perspectives. The perspectives included in the *BABOK® Guide* are:

- Agile,
- Business Intelligence,
- Information Technology,
- Business Architecture, and
- Business Process Management.

These perspectives do not presume to represent all the possible perspectives from which business analysis is practiced. The perspectives discussed in the *BABOK® Guide* represent some of the most common views of business analysis at the time of writing.

Any given initiative includes one, many, or all of these perspectives. For example, an initiative may have a technology component (Information Technology Perspective), the technology component may mean business process changes (Business Process Management Perspective), the initiative may decide to do part, or all of the work with an agile approach (Agile Perspective). Another initiative may merge two organizations and need to look at the business capabilities and how the transformation impacts those capabilities (Business Architecture Perspective), and the business leaders need updated information for decision making and analysis (Business Intelligence Perspective). Large or complex initiatives will likely employ all perspectives.

While the business analysis tasks detailed in the *BABOK® Guide* are intended to be applicable across all areas of business analysis, they are also pertinent to each specific business analysis perspective. Perspectives provide ways to approach business analysis work in a more focused manner suitable to the context. The perspectives help interpret and understand the knowledge areas and tasks in the *BABOK® Guide* from the lens in which one is currently working.

Each perspective follows a common structure:

- Change Scope,
- Business Analysis Scope,
- Methodologies, Approaches, and Techniques,
- Underlying Competencies, and
- Impact on Knowledge Areas.

11.1 The Agile Perspective

The Agile Perspective highlights the unique characteristics of business analysis when practiced in the context of agile environments.

Agile is about having a flexible mindset, embodied in a set of values and principles and exhibited by a variety of complementary practices. Agile initiatives involve constant change. Business analysts working on agile initiatives continually reassess, adapt, and adjust their efforts and tactics. Business analysts conduct analysis and deliver work products at the last responsible moment to continually allow flexibility for change; detailed analysis work is not done ahead of time, but just in time to be effectively utilized by the agile team.

Agile business analysis ensures that information is available to the agile team at the right level of detail at the right time. Business analysts help agile teams answer these questions:

- What need are we trying to satisfy?
- Is that need worth satisfying?
- Should we deliver something to satisfy that need?
- What is the right thing to do to deliver that need?

Business analysis work is performed continuously throughout an agile initiative and relies heavily on interpersonal skills such as communication, facilitation, coaching, and negotiation. Business analysts are active members of an agile team and often facilitate planning, analyzing, testing, and demonstrating activities. In an agile team, business analysis may be performed by a product manager/owner, business analyst, or by other defined team roles. Business analysts help the team identify modifications in assumptions and other project variations that emerge.

Refer to the *Agile Extension to the BABOK® Guide* for an expanded treatment of the role, mindset, and practices of business analysis in agile approaches, as well as details on the values and principles of the Agile Manifesto (www.agilemanifesto.org).

11.1.1

Change Scope

Business analysts working on agile initiatives engage with the business sponsor on a strategic level and assist with defining how the proposed product or feature aligns with the organization's objectives. They collaborate with various stakeholders and the change team to break the product vision down into a prioritized list of desired work items to be completed. The prioritized items (or prioritized backlog list) usually focus on the capabilities needed in the resultant product, with emphasis on the highest value items first.

Business analysts may act as a stakeholder proxy, or work directly with the sponsor or product owner.

In agile environments, change and rapid response to change is expected. Agile teams deliver small, incremental changes and commit to prioritized work items for only one iteration at a time. This allows the agile team to handle emerging changes for the upcoming iteration with minimal impact. An iteration is an agreed period of work time.

Requirements are developed through continual exploration and analysis of the business needs. It is important to note that though most agile approaches are iterative, not all iterative approaches are agile. There are also several agile approaches that are not iterative, such as the kanban method.

During agile initiatives, scope is constantly evolving. This is managed by the backlog list which is continually reviewed and re-prioritized. This process contributes to the refinement and redefinition of scope in order to meet the evolving and emerging business need.

If a major change emerges that significantly impacts the overall value and goals for the project, the project can be adjourned and reassessed.

.1 Breadth of Change

Agile approaches are used to address a variety of needs in an enterprise. The most common use of agile practices is in software development projects. However, many organizations have started to apply agile principles to non-software related change such as process engineering and business improvement. Initiatives using agile approaches can be undertaken within a single department or can span across multiple teams, departments, and divisions of an organization.

For organizations new to the agile mindset and practices, a focus on continuous improvement, ongoing changing behaviour, and making progress enables the organization to move towards culturally adopting the agile mindset. Adopting the agile mindset refers to the cultural adoption of agile principles as opposed to the organization considering agile as a methodology or practice to be implemented.

.2 Depth of Change

Initiatives using an agile approach are frequently part of a larger program of work, which can include organizational transformation and change, business process re-engineering, or business process change. The agile work stream is frequently, but not always, centered on software development. The other

elements of the program can be developed using agile or another methodology that is appropriate for the need. Agile principles and practices are often successfully applied in initiatives where:

- there is a clear commitment from the customer and engagement by empowered subject matter experts (SMEs),
- the business need or proposed solution is complex or complicated, and
- business needs are changing or unknown and are still emerging.

Agile approaches can be used for initiatives that are developing a solution for the first time, or for maintaining and enhancing an existing solution. For example, if the change is mission critical then processes can be added to address regulatory requirements and to deal with the mission critical aspects of the project.

.3 Value and Solutions Delivered

The value and solutions delivered in an agile initiative are similar to any other initiative. The difference with an agile approach is the emphasis on delivering value early in a highly collaborative manner, using adaptive planning that has a focus on continuous improvement.

An agile initiative provides value by virtue of the approach taken by an agile team through ongoing review and feedback of the work performed. Stakeholders get the opportunity to frequently review the product, which allows them to identify any missed requirements early. The solution evolves over time with an expectation of rapid and flexible response to change. Clarity and visibility of all communications is of the utmost importance to ensure the agile team's efforts align with the organization's needs and expectations.

In a new team, the business analyst often plays a central role in building rapport and trust amongst the agile team members and external stakeholders to help enable ongoing collaborative discussions and engagement. This interaction enables the agile team to accurately deliver value that meets evolving stakeholder needs.

.4 Delivery Approach

Agile approaches focus on people interactions, transparent communications, and ongoing delivery of valuable change to stakeholders.

Each agile approach has its own unique set of characteristics that allows teams to select an approach that best suits the initiative at hand. Some agile teams have found that a hybrid or combination of approaches is necessary to work within the constraints of their environment.

Refer to the *Agile Extension to the BABOK® Guide* for a description of different agile delivery approaches.

.5 Major Assumptions

The assumptions in place in agile environments frequently include:

- Changing requirements are welcome, even late in development.

- The business problem can be reduced to a set of needs that can be met using some combination of technology and business process change.
- Agile initiatives have fully engaged customers and empowered SMEs with complete buy-in to the agile approach.
- Ideally, team membership is constant and members are not continually being moved to other teams.
- There is a preference for multidisciplinary and co-located teams encouraging more efficient and effective face-to-face conversation. However, agile approaches can work well with distributed teams provided appropriate support and communication channels are in place.
- Team members may perform more than one role within the team if it is required, and provided that the team has the appropriate skills (for example, cross-functional teams).
- Team members have a mindset for continuous improvement and successful value delivery through regular inspection.
- Agile teams are empowered and self-organizing.

11.1.2 Business Analysis Scope

.1 Change Sponsor

It is important that a sponsor of an agile initiative be familiar with the agile philosophy, mindset, and approaches, and also be open to the constant feedback that will require trade-offs from the stakeholders.

An agile sponsor understands and accepts the:

- use of adaptive planning over predictive planning,
- use and value of a fixed period of time for a work cycle, and
- need and value of the sponsor's involvement.

The sponsor's (or empowered SME's) active involvement with the agile team is critical to providing the sponsor with the ability to preview and understand the product being developed, as well as allowing an opportunity for the sponsor to provide continuous feedback to the team and adjust the product as needs change.

.2 Change Targets and Agents

Agile approaches are most successful when the organizational culture and working environments lend themselves to intensive collaboration, frequent communication, and a strong disposition towards incremental delivery of appropriate solution value.

Agile teams are frequently either small or around teams of small teams. The simpler and flatter structure doesn't change the fact that the deliverables may affect a large group of stakeholders. The change agent, also considered a

stakeholder, is not different because the project uses agile. The primary agents for a change using an agile approach can include:

- **Agile team leader:** the facilitator of the work of the team. An agile team leader frequently shares the same soft skill set of a project manager, but completely delegates the tasks of planning, scheduling, and prioritization to the team. Rather than traditional command-and-control management, servant leadership is preferred in all the agile approaches. Depending on the approach, this role may be called scrum master, iteration manager, team leader, or coach.
- **Customer representative or product owner:** the active team member responsible for ensuring that the change being developed addresses the requirements for which it has been mandated. In Scrum this role is called the product owner. The dynamic systems development method (DSDM) refers to this role as that of a visionary, and extreme programming (XP) refers to it as a customer representative.
- **Team members:** the specialists or domain experts that include both technical and customer representation. Depending on size and particular context of the initiative, individuals within a team have different specialties. Usability experts, technical architects, and database administrators are just a sample of such specialized roles that provide support to the team as needed.
- **External stakeholders:** all of the remaining stakeholders who may not be considered team members, but are an interested party in the outcome of the project or simply required for its completion, playing what can be considered a supporting role in the team.

.3 Business Analyst Position

An agile team may have one or more team members with business analysis skills who may or may not have the job title of business analyst. This recognition of cross-skilled team members expands the practice of business analysis beyond that of a single specialist role.

On agile teams, business analysis activities can be performed by one or a combination of:

- a business analyst working on the team,
- the customer representative or product owner, or
- distributing these activities throughout the team.

Refer to the *Agile Extension to the BABOK® Guide* for more details.

.4 Business Analysis Outcomes

In an agile environment, business analysis brings people together and ensures that the right stakeholders are involved with the agile team at the right time. Open communication and collaboration is one of the principal outcomes of successful business analysis in an agile project.

Business analysts ensure that the project's vision and direction are in strategic alignment to the organizational goals and business need. The business analyst holds shared responsibility in defining strategic criteria for project completion and during the project assists with defining acceptance criteria. They also facilitate the articulation of the product vision statement. The product vision statement is a common initial deliverable.

Documentation rigour and style is highly dependent on the purpose and the context in which it is produced. Agile approaches favour just enough and just-in-time documentation rather than establishing predefined models for documentation to be delivered. This documentation approach allows for the documents to incorporate as much of the change introduced as possible while keeping the cost of change low. Mandatory documentation, such as that required for auditing or compliance reporting, are still produced as part of each delivery cycle. It is important that documents address an identified need and deliver more value than the cost incurred to produce and maintain them.

11.1.3

Approaches and Techniques

.1 Approaches

Agile is an umbrella term for a variety of approaches. All agile approaches practice business analysis but only a few explicitly define the business analysis role. The primary characteristic of any agile approach is its alignment to the values and principles of the Agile Manifesto. An agile team may implement or evolve to use a combination of approaches which enables them to deliver value more effectively given their project type and work environment.

Table 11.1.1: Agile Approaches

Approach	Brief description
Crystal Clear	Part of a family of Crystal methodologies which are defined based on hardness and colour. The hardness refers to the business criticality or potential for causing harm, which amounts to more rigour and predictive planning being required as the criticality increases. Colour refers to the heaviness of the project across a number of dimensions including number of people required and risk elements in the project.
Disciplined Agile Delivery (DAD)	A decision process framework which incorporates ideas from a variety of other agile approaches. It is intended to support a project from initiation through delivery. DAD is not prescriptive and allows for teams to customize their own life cycles and approaches.

Table 11.1.1: Agile Approaches (Continued)

Approach	Brief description
Dynamic Systems Development Method (DSDM)	A project delivery framework which focuses on fixing cost, quality, and time at the beginning while contingency is managed by varying the features to be delivered. MoSCoW prioritization technique is used for scope management. Time boxes, or short focused periods of time with clearly defined outcomes, are used to manage the work.
Evolutionary Project Management (Evo)	A project management method for developing and delivering a system incrementally. It has a strong focus on quantifying value for multiple stakeholders and planning increments based on delivery of that value (which can be measured). It uses impact estimation tables as a formal technique for assessing solutions for their ability to deliver value to multiple stakeholders for a given cost.
Extreme Programming (XP)	Named for the concept of taking beneficial software engineering techniques to the extreme. This concept focuses on the technical development processes and features pair-programming, test-driven development, and other craftsmanship approaches to the technical practices. XP technical practices are often used in conjunction with one of the agile management frameworks.
Feature Driven Development (FDD)	Focuses on a client valued functionality perspective to develop working software. For example, following a high-level scoping exercise, a feature list is identified and all planning, design, and development are performed based on feature sets.
Kanban	Does not require fixed iterations. Work moves through the development process as a continuous flow of activity. A key feature is to limit the amount of work underway at any one time (referred to as the work in progress limit or WIP). The team works only on a fixed number of items at any one time and work may begin on a new item only when it is required to maintain flow downstream and after the previous item has been completed.
Scaled Agile Framework® (SAFe™)	A framework for implementing agile practices at enterprise scale. It highlights the individual roles, teams, activities and artifacts necessary to scale agile from the team to program to the enterprise level.
Scrum	A lightweight process management framework based on empirical process control. Work is performed in a series of fixed length iterations, called Sprints, which last one month or less. At the end of each sprint the team must produce working software of a high enough quality that it could potentially be shipped or otherwise delivered to a customer.

.2 Techniques

The following table lists techniques commonly used within agile approaches. Refer to the *Agile Extension to the BABOK® Guide* for a more detailed description of these techniques.

Table 11.1.2: Techniques used within Agile Approaches

Technique	Brief Description
Behaviour Driven Development (BDD)	An approach that enhances the communication between stakeholders and team members by expressing product needs as concrete examples.
Kano Analysis	A technique for understanding which product features will help drive customer satisfaction.
Lightweight Documentation	A principle that governs all documentation produced on an agile project. The purpose is to ensure that all documentation is intended to fulfill an impending need, has clear value for stakeholders, and does not create unnecessary overhead. For example, a system overview document may be written towards the end of a project based on stable content and acceptance tests written as part of the product testing.
MoSCoW Prioritization	A method to prioritize stories (or other elements) in incremental and iterative approaches. MoSCoW (must have, should have, could have, won't have) provides a way to reach a common understanding on relative importance of delivering a story or other piece of value in the product.
Personas	Fictional characters or archetypes that exemplify the way that typical users interact with a product.
Planning Workshop	A collaborative workshop that is used to allow an agile team to determine what value can be delivered over a time period such as a release.
Purpose Alignment Model	A model that is used to assess ideas in the context of customer and value.
Real Options	An approach to help people know when to make decisions rather than how.
Relative Estimation	Team estimation techniques using either story points, which represent the relative complexity of a user story to develop, or ideal days, which represent the amount of total effort a story would take to develop.
Retrospectives	A similar term for the Lesson Learned technique. Retrospectives focus on continuous improvement of the teamwork process and are held after every iteration on agile projects.

Table 11.1.2: Techniques used within Agile Approaches (Continued)

Technique	Brief Description
Story Decomposition	Ensures that the requirements for a product are represented at the appropriate level of detail and are derived from a valuable business objective.
Story Mapping	Provides a visual and physical view of the sequence of activities to be supported by a solution.
Storyboarding	Detail visually and textually the sequence of activities that represent user interactions with a system or business.
Value Stream Mapping	Provides a complete, fact-based, time-series representation of the stream of activities required to deliver a product or service to the customer.

11.1.4 Underlying Competencies

Agile is a mindset. Agile business analysts embody the values and principles of the Agile Manifesto which are based on a humanistic view of product development as a process founded in communication and collaboration. Refer to the *Agile Extension to the BABOK® Guide* for a description of the principles for business analysts. In adopting the agile mindset and philosophy, the business analyst develops competencies in:

- **Communication and collaboration:** the ability to communicate the sponsor's vision and needs; assist in influencing others to support the vision; participate and possibly facilitate negotiation of priorities; and facilitate collaborative agreement on solution outcomes.
- **Patience and tolerance:** the ability to maintain self-control under pressure and keep an open mind when interacting with others.
- **Flexibility and adaptability:** cross-functional skill sets that allow the business analyst to step outside their specialization in order to support other team members.
- **Ability to handle change:** the ability to quickly assess the impact of change and determine what provides business value amongst frequently changing requirements, and assisting with, or maintaining, the re-prioritization of the to-do work list.
- **Ability to recognize business value:** the ability to understand how changes and new features can achieve business value and support the vision.
- **Continuous improvement:** periodically review with the agile team how to become more effective.

11.1.5 Impact on Knowledge Areas

This section explains how specific business analysis practices within agile are mapped to business analysis tasks and practices as defined by the *BABOK® Guide*. It also describes how each knowledge area is applied or modified with the agile discipline.

Each knowledge area lists techniques relevant to an agile perspective. *BABOK® Guide* techniques are found in the Techniques chapter of the *BABOK® Guide*. Agile Extension techniques are discussed in detail in the *Agile Extension to the BABOK® Guide*. This is not intended to be an exhaustive list of techniques but rather to highlight the types of techniques used by business analysts while performing the tasks within the knowledge area.

.1 Business Analysis Planning and Monitoring

In agile approaches, detailed business analysis planning can be deferred until work on an activity is ready to begin rather than done upfront as in predictive projects.

An initial plan for business analysis activities is developed at the beginning of the project. The plan then gets updated prior to the start of each cycle to account for change and to ensure that the plan is always up to date. Stakeholder involvement and engagement is key to the success of agile projects. Business analysts proactively plan to involve, engage, and collaborate with stakeholders. Communication is commonly much less formal and business analysis deliverables are often interactions and collaboration with less emphasis on the written documents.

BABOK® Guide Techniques

- Backlog Management (p. 220)
- Collaborative Games (p. 243)
- Estimation (p. 271)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Mind Mapping (p. 299)
- Prioritization (p. 311)
- Scope Modelling (p. 338)
- Stakeholder List, Map, or Personas (p. 344)
- User Stories (p. 359)
- Workshops (p. 363)

Agile Extension Techniques

- Lightweight Documentation
- MoSCoW Prioritization
- Personas
- Relative Estimation
- Retrospective

.2 Elicitation and Collaboration

Progressive elicitation and elaboration occur throughout an agile initiative. The most common pattern is an initial elicitation activity that establishes the high-level vision and scope of the solution, and an initial milestone-based plan for the delivery of the product. In every cycle there is more detailed elicitation for the

backlog items that will be developed in that cycle. The intent of elicitation activities is to generate just enough detail to ensure that the work at hand is performed correctly while aiming towards the goals. Agile approaches aim to minimize the time between the elaboration of needs and their implementation in the solution. There is a strong focus on collaborative elicitation approaches such as workshops with stakeholders.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Backlog Management (p. 220)
- Brainstorming (p. 227)
- Collaborative Games (p. 243)
- Concept Modelling (p. 245)
- Interface Analysis (p. 287)
- Mind Mapping (p. 299)
- Non-Functional Requirements Analysis (p. 302)
- Process Modelling (p. 318)
- Prototyping (p. 323)
- Reviews (p. 326)
- Scope Modelling (p. 338)
- Stakeholder List, Map, or Personas (p. 344)
- Use Cases and Scenarios (p. 356)
- User Stories (p. 359)
- Workshops (p. 363)

Agile Extension Techniques

- Behaviour Driven Development
- Lightweight Documentation
- Personas
- Storyboarding
- Story Mapping

.3 Requirements Life Cycle Management

As agile initiatives unfold, the scope is defined with increasing specificity. The expectation is that the needs will change and that the design will evolve over the course of the project. Prioritization of features based on value and development priority drives the work done in each cycle. Validation of the evolving solution with the stakeholders occurs at the end of every iteration in place of a formal requirements approval process.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Backlog Management (p. 220)
- Collaborative Games (p. 243)
- Prioritization (p. 311)
- Reviews (p. 326)
- Workshops (p. 363)

Agile Extension Techniques

- Kano Analysis
- MoSCoW Prioritization
- Story Decomposition
- Story Mapping

.4 Strategy Analysis

Agile approaches are often used when there is uncertainty about the needs, the solution, or the scope of change. Strategy analysis is a constant part of an agile initiative to ensure that the solution delivered continues to provide value to stakeholders. Agile team members use strategy analysis to help understand and define product vision, and develop and adjust the development roadmap, in addition to conducting ongoing assessments of related risks. For every iteration, the proposed solution is reassessed against the current business context to ensure that it will effectively meet the business goals. The adaptive nature of agile projects means that adapting the project to changes in the organization's goals is not disruptive; rather, it is an expected part of the process.

BABOK® Guide Techniques

- Backlog Management (p. 220)
- Brainstorming (p. 227)
- Business Capability Analysis (p. 230)
- Collaborative Games (p. 243)
- Concept Modelling (p. 245)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Scope Modelling (p. 338)
- Workshops (p. 363)

Agile Extension Techniques

- Kano Analysis
- Personas
- Purpose Alignment Model
- Real Options
- Value Stream Analysis

.5 Requirements Analysis and Design Definition

Needs are progressively elaborated during an agile project. Analysis and design are performed on a just-in-time basis, either just before or during the iteration in which the solution component will be developed.

Analysis performed just before the iteration is to provide the team with enough information to estimate the planned work. Analysis performed during the iteration is to provide the team with enough information to construct or deliver the planned work.

Models and other analysis and design techniques are typically used informally, and may not be maintained once they have served their purposes. The analysis and design approach used should support progressive elaboration, be adaptable to change based on learning, and not cause the team to select solutions prematurely. Agile teams tend to use user stories at the lowest level of decomposition, usually supported by acceptance criteria which capture the analysis and design details regarding how the stories should behave when implemented. Validation of the evolving solution is performed with stakeholders at the end of every iteration.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Business Capability Analysis (p. 230)
- Business Rules Analysis (p. 240)

- Collaborative Games (p. 243)
- Concept Modelling (p. 245)
- Interface Analysis (p. 287)
- Non-Functional Requirements Analysis (p. 302)
- Prioritization (p. 311)
- Process Analysis (p. 314)
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Agile Extension Techniques

- Behaviour Driven Development
- Kano Analysis
- Lightweight Documentation
- MoSCoW Prioritization
- Purpose Alignment Model
- Real Options
- Story Decomposition
- Story Elaboration
- Story Mapping
- Storyboarding
- Value Stream Analysis

.6 Solution Evaluation

Throughout an agile project, the stakeholders and agile team continually assess and evaluate the development solution as it is incrementally built and refined. Evaluation of the evolving solution with the stakeholders occurs at the end of every development cycle to ensure the deliverable meets their needs and satisfies their expectations. The business analyst ensures that the product meets expectations before a product is released, and identifies new opportunities that will add value to the business.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Business Capability Analysis (p. 230)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Non-Functional Requirements Analysis (p. 302)
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- Reviews (p. 326)
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- Use Cases and Scenarios (p. 356)
- User Stories (p. 359)
- Workshops (p. 363)

Agile Extension Techniques

- Personas
- Value Stream Analysis

11.2 The Business Intelligence Perspective

The Business Intelligence Perspective highlights the unique characteristics of business analysis when practiced in the context of transforming, integrating, and enhancing data.

The focus of business intelligence is the transformation of data into value-added information: where to source it, how to integrate it, and how to enhance and deliver it as analytic insight to support business decision making.

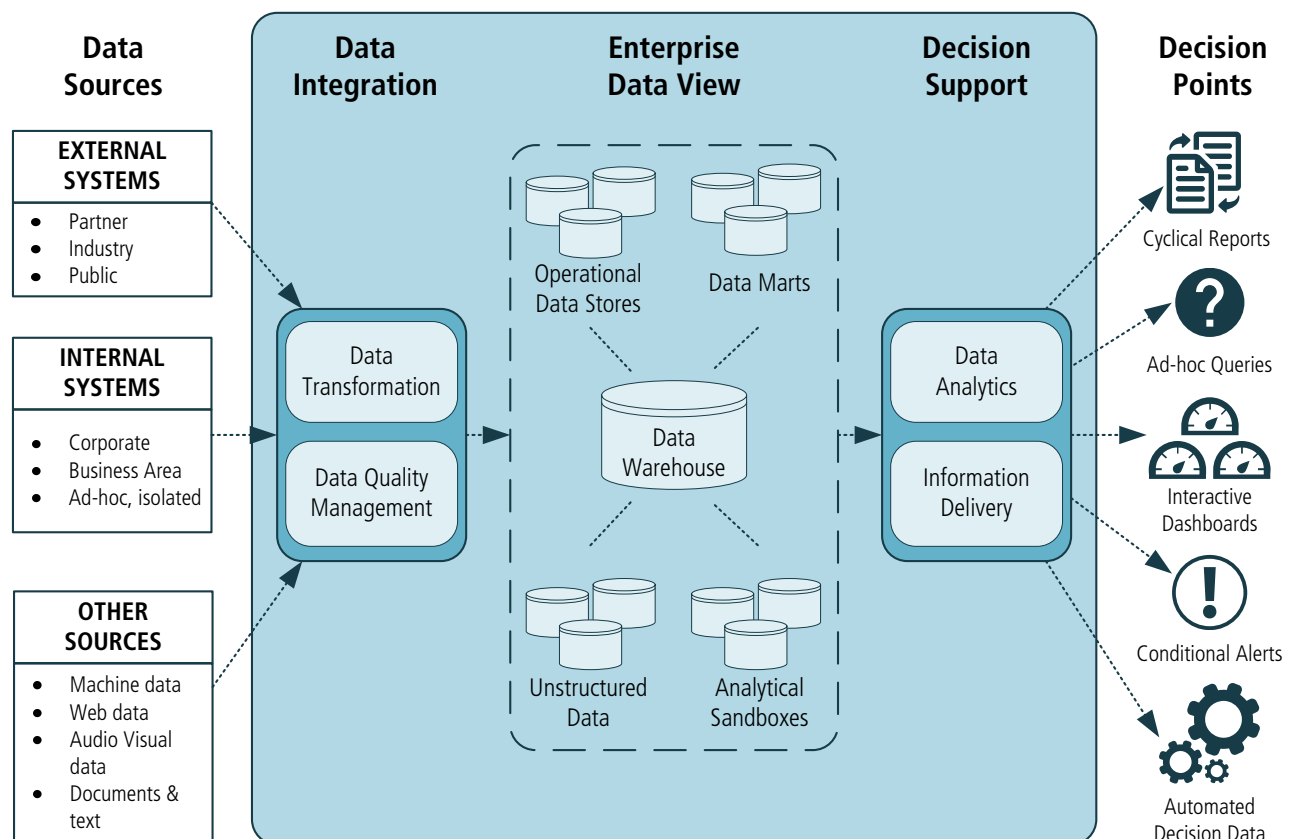
Business intelligence initiatives apply data-centric system architectures as well as technologies and tools to deliver reliable, consistent, high-quality information that enables stakeholders to better manage strategic, tactical, and operational performance.

11.2.1 Change Scope

.1 Breadth of Change

A key objective of a business intelligence system is the consistent definition and usage of information throughout an organization by establishing a 'single point of truth' for diverse business data. A solution architecture that can integrate multiple data sources from within (and potentially from outside) the organization provides the foundation of a business intelligence solution.

Figure 11.2.1: Business Intelligence Solution - Conceptual Framework



The business intelligence promotes an enterprise-wide view of information management. To support that conceptual framework, a business intelligence initiative may also involve the development of infrastructure services in the organization, such as data governance and metadata management.

.2 Depth of Change

Business intelligence initiatives focus on the information needed to support decision making at, or across, different levels within the organization:

- **executive level:** supports strategic decisions,
- **management level:** supports tactical decisions, or
- **process level:** supports operational decisions.

Where information needs are initially expressed or identified at a particular level, the business analyst investigates the business implications at other levels to assess the overall impact of the change on the organization.

At each level, the business needs may involve any or all of the following:

- communication requirements for the development of new reporting or the replacement of existing reporting,
- information requirements for the addition or extension of analytic functionality, and/or
- data integration requirements for the construction or modification of the enterprise data view with regard to data sources, definitions, transformation rules and quality issues.

.3 Value and Solutions Delivered

The value of a business intelligence initiative is in its ability to provide timely, accurate, high value, and actionable information to those people and systems who can use it effectively in making business decisions.

Better informed decision making at all levels can lead to improved business performance in:

- strategic processes such as market analysis, customer engagement, and product development,
- tactical processes such as stock control and financial planning, and
- operational processes such as credit assessment, fault detection, and accounts payable monitoring.

These improvements in an organization's current and future performance may be realized as increased revenues and reduced costs.

.4 Delivery Approach

A business intelligence solution presents a range of delivery options to meet the emerging information needs of stakeholders and the priorities of the organization.

The extensibility and scalability of the solution architecture provide for the support of business decision making to be progressively introduced or enhanced:

- at different levels in the organization, from strategic (senior executive), through tactical (management), to operational (staff and systems), and
- in target functional areas in the organization, from a specific area through to an enterprise-wide implementation.

The infrastructure services that provide data management, analytics, and presentation capabilities, facilitate a phased or incremental development strategy in respect of:

- the inclusion, coordination and control of different data sources, and
- the analysis and development of business information and insights.

Infrastructure components of a business intelligence solution are often provided by a commercial off-the-shelf package configured to the specific business environment and needs.

.5 Major Assumptions

The following is a list of major assumptions of a business intelligence initiative:

- existing business processes and transactional systems can provide source data that is definable and predictable,
- the cross-functional data infrastructure that is needed to support a business intelligence solution has not been precluded by the organization on technical, financial, political/cultural, or other grounds, and
- the organization recognizes that process re-engineering and change management might be needed in order to effectively realize the value from a business intelligence solution.

11.2.2 Business Analysis Scope

.1 Change Sponsor

The change sponsor of a business intelligence initiative is ideally the highest level role from the organizational unit affected by the change. This provides for a consistent, cohesive approach to the shared usage of data assets within the cross-functional architecture of a business intelligence solution.

.2 Change Targets

The targets of a business intelligence initiative are the business decisions made by people or processes at multiple levels in the organization that can be improved by better reporting, monitoring, or predictive modelling of performance-related data.

.3 Business Analyst Position

As in other initiatives, the business analyst acts as the primary liaison between business intelligence stakeholders and solution providers in the elicitation,

analysis, and specification of business needs.

In addition to that role, the business analyst may also participate in technical activities that are specific to business intelligence, including:

- enterprise data modelling,
- decision modelling,
- specialized presentation design (for example, dashboards), and
- ad hoc query design.

A business analyst working on a business intelligence initiative serves in one or in a combination of the following roles:

- business analyst who is competent in the definition of business requirements and the assessment of potential solutions,
- business intelligence functional analyst who has an understanding of data mining and predictive analytic techniques, as well as skills in developing visualizations,
- data analyst who is experienced at defining source systems data to be used for the required analytical purposes, or
- data modeller/architect who is skilled in defining the source and target data structures in logical data models.

.4 Business Analysis Outcomes

In the business intelligence discipline, business analysis is focused on the major components of the solution architecture:

- the specification of business decisions to be influenced or changed,
- the collection of data from source systems,
- the integration of divergent sources into a convergent enterprise framework, and
- the provision of targeted information and analytic insight to business stakeholders.

The business analyst is responsible for the analysis and specification of the business requirements for all of these components and collaborates with technical specialists to assess solution artifacts.

The major outcomes of business analysis are:

- **Business process coverage:** defines the scope of the change with a high-level overview of the business decisions within the enterprise that are to be supported by the solution. It identifies how the information output will be used and what value it will provide.
- **Decision models:** identify the information requirements of each business decision to be supported and specify the business rules logic of how the individual information components contribute to the decision outcome.
- **Source logical data model and data dictionary:** the source logical data model provides a standard definition of the required data as held in each

source system. The source data dictionary provides a definition of each element and the business rules applied to it: business description, type, format and length, legal values, and any inter-dependencies.

- **Source data quality assessment:** evaluates the completeness, validity, and reliability of the data from source systems. It identifies where further verification and enhancement of source data is required to ensure consistent business definitions and rules apply across the enterprise-wide data asset.
- **Target logical data model and data dictionary:** the target logical data model presents an integrated, normalized view of the data structures required to support the business domain. The target data dictionary provides the standardized enterprise-wide definition of data elements and integrity rules.
- **Transformation rules:** map source and target data elements to specify requirements for the decoding/encoding of values and for data correction (error values) and enrichment (missing values) in the transformation process.
- **Business analytics requirements:** define the information and communication requirements for decision support outputs. These include:
 - predefined reports,
 - dashboards,
 - balanced scorecards,
 - ad hoc reports,
 - online analytical processing (OLAP) queries,
 - data mining,
 - prescriptive analytics,
 - conditional alerts,
 - complex event processing, and
 - predictive modelling.
- Specifications for each output can include: (1) data selections/dimensions, level of granularity, filtering criterion applied, possibilities for drill down, slice and dice, and user access and permissions; and (2) presentation rules to define data element format, translation (labels, look-ups), calculations, and data aggregations.
- **Solution architecture:** provides a high-level design view of how the decision support requirements of each functional area will map to the business intelligence framework. It is typically presented in the form of a process (or data flow) model that defines:
 - where the source data is held,
 - how (pull/push) and when (frequency, latency) the data will be extracted,

- where the transformations will take place (cleansing, encoding, enhancement),
- where the data will be physically stored (data warehouse, data marts), and
- how the data will flow to presentation outputs (reporting facilities, query tools).

11.2.3 Methodologies and Approaches

.1 Methodologies

There are no formalized business intelligence methodologies that impact the responsibilities and activities of the business analyst. However, a business intelligence initiative can operate within or alongside methodologies applicable to other disciplines or perspectives which themselves might impact the business analysis role.

.2 Approaches

Within the business intelligence framework there are a number of less formal and potentially overlapping approaches that map to particular business and technical contexts.

Types of Analytics

There are three types of data analytics that represent incremental solutions, with increasing levels of systems complexity, cost, and value:

- **Descriptive analytics:** uses historical data to understand and analyze past business performance. Business information can be categorized and consolidated to best suit the stakeholder's view including executive management dashboards, middle level management key performance indicator (KPI) scorecards, and operational level management charts. No assumptions are made as to which situations are of interest to the stakeholders, what decisions need to be made, or what actions might be carried out. The business analysis focus is on the information and communication requirements for standard reporting and dashboards, ad hoc reporting, and query functionality.
- **Predictive analytics:** applies statistical analysis methods to historical data to identify patterns, and then uses that understanding of relationships and trends to make predictions about future events. The particular situations that are of interest to the stakeholders are specified, and their business rules are defined. The business analysis focus is on the information requirements for pattern recognition through data mining, predictive modelling, forecasting, and condition-driven alerts.
- **Prescriptive analytics:** expands on predictive analytics to identify decisions to be made and to initiate appropriate action to improve business performance. Statistical optimization and simulation techniques can be

used to determine the best solution or outcome among various choices. For situations of interest to stakeholders, full specification of the associated decisions and potential actions are required. The business analysis focus is on the business objectives, constraints criteria, and the business rules that underpin the decision-making process.

Supply and Demand Driven

The objectives and priorities of a business intelligence initiative can be based on the technical goals of improving existing information delivery systems (supply-driven) or on the business goals of providing the appropriate information to improve decision-making processes (demand-driven):

- **Supply-driven:** assumes the view of "for a given cost, what value can we deliver?". This approach maps existing systems data to define what data is available. A common implementation strategy would be to:
 1. phase the inclusion of existing databases into the business intelligence solution architecture,
 2. progressively replace or repair existing outputs, and
 3. explore new insights that might be gained from the consolidated data.
- **Demand-driven:** assumes the view of "for a given value, what cost do we incur?". This approach starts with identifying the information output needed to support business decisions, and then tracing that information back to the underlying data sources to determine feasibility and cost. It provides for incremental implementation strategies that are not determined by existing database structures, and allows for early exploratory usage of business intelligence beyond existing reporting requirements.

Structured and Unstructured Data

There are two types of data that business intelligence approaches consider:

- **Structured data:** traditional data warehouse solutions have been based on consolidating the structured data (numerical and categorical) recorded in operational systems where business information sets are identified by predefined structures (referred to as 'schema on write') and where a rules-driven template ensures data integrity. The business analysis focus is on data models, data dictionaries, and business rules to define information requirements and capabilities.
- **Unstructured data:** business intelligence solutions can include semi-structured or unstructured data which includes text, images, audio, and video. This data frequently comes from external sources. For this type of data, the structure and relationships are not predefined and no specific organization rules have been applied to ensure data integrity. Information sets are derived from the raw data (referred to as 'schema on read'). The business analysis focus is on metadata definitions and data matching algorithms to define information requirements and capabilities.

11.2.4 Underlying Competencies

As in any business analysis discipline, the business analyst requires the fundamental communication and analytical competencies to be effective in liaising with both business stakeholders and technical solution providers.

In the business intelligence discipline, this coordination of business information requirements with business intelligence systems outcomes can be further enhanced by the business analyst's specific competencies in:

- business data and functional usage, including terminology and rules,
- the analysis of complex data structures and their translation into standardized format,
- business processes affected including KPIs and metrics,
- decision modelling,
- data analysis techniques including basic statistics, data profiling, and pivoting,
- data warehouse and business intelligence concepts and architecture,
- logical and physical data models,
- ETL (Extract, Transform, Load) best practices including historical data track and reference data management, and
- business intelligence reporting tools.

11.2.5 Impact on Knowledge Areas

This section explains how specific business analysis practices within business intelligence are mapped to business analysis tasks and practices as defined by the *BABOK® Guide*. This section describes how each knowledge area is applied or modified with the business intelligence discipline.

Each knowledge area lists techniques relevant to a business intelligence perspective. Techniques used in the discipline of business intelligence do not deviate, to any great extent, from the *BABOK® Guide* techniques. *BABOK® Guide* techniques are found in the Techniques chapter of the *BABOK® Guide*. This is not intended to be an exhaustive list of techniques but rather to highlight the types of techniques used by business analysts while performing the tasks within the knowledge area.

.1 Business Analysis Planning and Monitoring

A business intelligence initiative may require establishing an underlying data infrastructure to support the solution, or it might be an enhancement based on the infrastructure of an existing solution. Scope Modelling is frequently used to differentiate between these alternatives and plan the relevant business analysis activities accordingly.

The business intelligence paradigm of information delivery might be a new,

unfamiliar approach for business stakeholders and for the business analysts themselves. In planning the initiative, the business analyst considers:

- how experienced the stakeholders are in expressing their information and communication requirements in the business intelligence context, and
- how skilled the business analysts are in interpreting those requirements into detailed specifications for business intelligence technical specialists.

Business intelligence solutions typically provide frameworks, tools, and techniques that can assist in requirements definition and solution modelling. The level of stakeholders' and business analysts' expertise in these can have an impact on the planned approach.

When assessing stakeholder attitudes towards the business intelligence initiative, the business analyst should be aware that an enterprise-wide business intelligence solution might not provide direct value to some operational stakeholders, but will deliver it elsewhere in the organization, and the flexibility and extensibility provided by the business intelligence infrastructure delivers longer-term strategic value that goes beyond short-term operational benefits.

A business intelligence solution that integrates multiple data sources typically engages many stakeholders with overlapping information requirements. Business analysts prepare for the analysis and synthesis of individual requirements into a set that is complete and cohesive without conflicts and redundancies.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Balanced Scorecard (p. 223)
- Brainstorming (p. 227)
- Decision Analysis (p. 261)
- Estimation (p. 271)
- Functional Decomposition (p. 283)
- Interviews (p. 290)
- Item Tracking (p. 294)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Non-Functional Requirements Analysis (p. 302)
- Organizational Modelling (p. 308)
- Prioritization (p. 311)
- Process Modelling (p. 318)
- Reviews (p. 326)
- Risk Analysis and Management (p. 329)
- Roles and Permissions Matrix (p. 333)
- Root Cause Analysis (p. 335)
- Scope Modelling (p. 338)
- Stakeholder List, Map, or Personas (p. 344)
- Survey or Questionnaire (p. 350)
- Use Cases and Scenarios (p. 356)
- User Stories (p. 359)
- Workshops (p. 363)

.2 Elicitation and Collaboration

The cross-functional nature of business intelligence typically requires business analysts to employ specialized documentation tools and techniques to elicit particular types of requirements from stakeholders, both business and technical.

Individual stakeholders may only possess partial knowledge and expertise regarding:

- the business decisions that need support,
- the data elements that support those business decisions,
- the data sourcing, transformation, and integration rules, and
- the presentation of the required information.

Interviews with individual stakeholders identify the information and analytic insight required to support their decision making. Workshops with stakeholders from across different functional areas of the business can help detect common, overlapping information requirements that would be better met with an integrated solution.

Data models and data dictionaries provide definitions of the structure and business rules of existing systems data. The business analyst assesses available documentation to identify incompleteness of a model or inconsistencies between models.

Process models that are extended to include data artifacts can help identify the data sources required at decision points. Decision models specify the data analytic requirements and business rules for decisions.

Commercial off-the-shelf packages of business intelligence functionality can provide the business analyst with a set of highly effective prototyping tools to elicit and clarify stakeholder information and communication requirements.

BABOK® Guide Techniques

- | | |
|-------------------------------------|---|
| • Brainstorming (p. 227) | • Item Tracking (p. 294) |
| • Document Analysis (p. 269) | • Observation (p. 305) |
| • Focus Groups (p. 279) | • Prototyping (p. 323) |
| • Functional Decomposition (p. 283) | • Workshops (p. 363) |
| • Glossary (p. 286) | • Stakeholder List, Map, or Personas (p. 344) |
| • Interface Analysis (p. 287) | • Survey or Questionnaire (p. 350) |
| • Interviews (p. 290) | |

.3 Requirements Life Cycle Management

The architectural nature of the business intelligence discipline requires establishing the infrastructure capabilities in the solution. This can introduce structural dependencies within the solution, particularly where delivery is phased,

that affect the prioritization of individual business needs. It is often possible to achieve efficiencies by implementing related requirements at the same time.

BABOK® Guide Techniques

- Item Tracking (p. 294)
- Organizational Modelling (p. 308)
- Prioritization (p. 311)
- Reviews (p. 326)
- Roles and Permissions Matrix (p. 333)
- Stakeholder List, Map, or Personas (p. 344)
- Workshops (p. 363)

.4 Strategy Analysis

Business analysts can use high-level conceptual data models to map the current state of corporate information, to identify information silos, and to assess their related problems and opportunities. Organization Modelling can be used to evaluate any current data management infrastructure, such as metadata management and data governance.

In defining the future state strategy, business analysts can use high-level models to map the architecture for data storage and for data conveyance and transformation:

- **Logical data models:** provide a static view of the solution architecture, representing the information portal that connects the sourcing of operational data inputs with the delivery of the business information outputs.
- **Data flow diagrams:** are commonly used to map the dynamic aspects of the solution (data-in-motion) and to identify other architectural constructs such as latency and accessibility.
- **Decision models:** are useful for defining how relevant business decisions are made and where and how data analytics can be effectively used to meet these needs.
- **Physical data models:** show the implementation environment including the data warehouse and data marts.

The extensible architecture provided by business intelligence solutions can support incremental implementation across different functional areas of the business. Business analysts can define change strategy options based on business needs and priorities, impact on the business operations, and the usability of existing infrastructure components.

BABOK® Guide Techniques

- Backlog Management (p. 220)
- Benchmarking and Market Analysis (p. 226)
- Brainstorming (p. 227)
- Business Rules Analysis (p. 240)
- Data Flow Diagrams (p. 250)
- Data Modelling (p. 256)
- Decision Analysis (p. 261)

- Decision Modelling (p. 265)
- Document Analysis (p. 269)
- Estimation (p. 271)
- Focus Groups (p. 279)
- Functional Decomposition (p. 283)
- Glossary (p. 286)
- Organizational Modelling (p. 308)
- Risk Analysis and Management (p. 329)
- Root Cause Analysis (p. 335)
- Stakeholder List, Map, or Personas (p. 344)
- SWOT Analysis (p. 353)

.5 Requirements Analysis and Design Definition

When modelling and specifying back office data capture and storage requirements, business analysts use specific data-oriented modelling techniques such as Data Modelling, Data Dictionary, Decision Modelling, and Business Rules Analysis.

Models of an existing system's data help to define data availability and identify redundancies, inconsistencies, and data quality issues. Where existing systems documentation is non-existent or out of date, reverse-engineered modelling can be a substantial component of work, and frequently requires collaboration with technical experts such as database administrators and application programmers.

A future state data model demonstrates how the source information is generically structured in the proposed solution. The overall transformation process is commonly modelled using Data Flow Diagrams to illustrate the management of latency and accessibility requirements in the solution. Business analysts define specific business rules for data integrity checking and for data transformation.

For modelling and specifying front office information outputs, business analysts:

- analyze existing reports to determine if they are candidates to be replaced or repaired with business intelligence outputs, and
- use business intelligence capabilities such as ad hoc queries, data mining, and complex event processing to identify and specify the content and format of new business intelligence outputs.

Business analysts are involved in assessing the capability of a proposed solution (typically a commercial off-the-shelf software package) in respect of the specified requirements. In the business intelligence context, these include functional requirements such as self-serve facilities, data analytics tools, data presentation tools, drill down capabilities, and non-functional requirements related to issues such as data quality, data latency, and query performance.

.6 BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Balanced Scorecard (p. 223)
- Business Rules Analysis (p. 240)
- Data Dictionary (p. 247)
- Data Flow Diagrams (p. 250)
- Data Modelling (p. 256)
- Decision Modelling (p. 265)

- Document Analysis (p. 269)
- Functional Decomposition (p. 283)
- Glossary (p. 286)
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- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Non-Functional Requirements Analysis (p. 302)
- Observation (p. 305)
- Organizational Modelling (p. 308)
- Prioritization (p. 311)
- Process Modelling (p. 318)
- Prototyping (p. 323)
- Reviews (p. 326)
- Scope Modelling (p. 338)
- Sequence Diagrams (p. 341)
- Stakeholder List, Map, or Personas (p. 344)
- State Modelling (p. 348)
- Use Cases and Scenarios (p. 356)
- Vendor Assessment (p. 361)

.7 Solution Evaluation

A common enterprise limitation with the introduction of a business intelligence solution is the under-utilization of the information resource and analytic functionality that the solution provides. Stakeholders who are not familiar with the capabilities of business intelligence might focus on simply replacing or repairing existing information outputs. Business analysts explore and evaluate opportunities for additional value that are enabled by a business intelligence solution.

.8 BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Balanced Scorecard (p. 223)
- Business Rules Analysis (p. 240)
- Data Flow Diagrams (p. 250)
- Data Modelling (p. 256)
- Decision Analysis (p. 261)
- Decision Modelling (p. 265)
- Estimation (p. 271)
- Focus Groups (p. 279)
- Functional Decomposition (p. 283)
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- Interviews (p. 290)
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- Risk Analysis and Management (p. 329)
- Stakeholder List, Map, or Personas (p. 344)
- Survey or Questionnaire (p. 350)
- SWOT Analysis (p. 353)
- Use Cases and Scenarios (p. 356)
- User Stories (p. 359)
- Vendor Assessment (p. 361)

11.3 The Information Technology Perspective

The Information Technology Perspective highlights the characteristics of business analysis when undertaken from the point of view of the impact of the change on information technology systems.

This perspective focuses on non-agile approaches to IT initiatives.

When working in the information technology (IT) discipline, business analysts deal with a wide range of complexity and scope of activities. Initiatives may be as small as minor bug fixes and enhancements, or as large as re-engineering the entire information technology infrastructure for an extended enterprise. Business analysts are called upon to work with this diverse level of knowledge and skills among stakeholders to deliver valuable solutions to their IT needs.

For information regarding agile approaches within information technology initiatives, see The Agile Perspective (p. 368).

Being able to effectively articulate the business' vision and needs to technical stakeholders is central to the success of a business analyst in the information technology discipline. Business analysts proactively collaborate with both the business stakeholders and development teams to ensure that needs are understood and aligned with organizational strategy. A business analyst frequently plays the role of the translator who helps business and technology stakeholders understand each other's needs, constraints, and context. The concept of solution design is appropriate in a technology context, and from the IT business analyst's point of view. However, the term 'design', when discussed within an IT setting, is generally assumed to mean 'technical design' or the utilization of technologies to solve business problems. Business analysts within an IT context define and elaborate solution requirements or participate in solution design with business stakeholders while maintaining a separation with technical design.

Important In IT contexts, the term 'design' has traditionally been reserved for solution or technical design performed by developers, IT architects, or solution architects. All work done by IT business analysts is covered by the term 'requirements', including concepts such as the definition and design of business processes, user interfaces, reports or other elements of the solution relevant to stakeholders outside of the implementation team. Business analysts working in this context may prefer the term 'solution requirements' instead of 'design' in order to maintain a clear separation of responsibility.

Business analysts working in an information technology environment consider their tasks in light of three key factors:

- **Solution impact:** the value and risk of the solution to the business.
- **Organizational maturity:** the formality and flexibility of the organizational change processes.
- **Change scope:** the breadth, depth, complexity, and context for the proposed change.

11.3.1 Change Scope

Changes to IT systems are initiated for several reasons.

Each of the following triggers can lead to an IT change:

- **Create a new organizational capability:** can be executed to transform the organization. These types of IT initiatives may drive the creation of larger programs to address non-IT changes, but are centered on a technology that alters the business environment.
- **Achieve an organizational objective by enhancing an existing capability:** is part of a change that meets a defined need. This may include changes to meet regulatory requirements or to enable business specific goals. These types of initiatives often modify an existing system but may also require implementation and integration of new systems.
- **Facilitate an operational improvement:** is undertaken to improve organizational efficiency or reduce organizational risk. The change scope, organizational maturity, and solution impact dictate whether these changes will be managed as a project, part of a continuous improvement effort, or as an enhancement.
- **Maintain an existing information technology system:** is undertaken to ensure smooth operation of an existing IT system. Depending on the scope of the change, maintenance may be managed as a project or a regularly scheduled activity. This may include technology driven changes such as a vendor discontinuing support of a technology, scheduled releases or upgrades to a purchased software package, or technical modifications required to support architecture strategy.
- **Repair a broken information technology system:** is undertaken when an IT system that is not performing as expected is changed to correct the dysfunction. The urgency of the repair is generally based on the level of disruption caused. In some cases the scope of the repair effort is very large, so the repair is managed as a project.

.1 Breadth of Change

Information technology initiatives may focus on a single system or on multiple systems which interact with each other. Some systems are developed and maintained in-house while others are commercial off-the-shelf (COTS) systems developed by an organization that is external to the group implementing the system. It is also possible that an external organization completes custom development, such as when development tasks are outsourced or contracted.

The scope of an IT initiative is often narrowly focused on software and hardware and a minimal set of systems, applications, or stakeholders. Larger initiatives may impact multiple user groups or systems, and often require collaboration with the extended enterprise. The implementation of COTS information technology systems may begin with a small or limited scope when the change is initiated, but after analysis is complete the scope is broader than originally anticipated. The business analysis approach for a COTS selection and implementation is approached differently than in-house development. These IT systems almost always require customization, integration, administration, and training. In some cases, the initiatives are limited to initial installation and implementation, or enhancements to an existing application. IT initiatives may also focus on a very specific technology solution such as what data is needed, how data is gathered,

how it is stored and accessed in order to support business transaction methods, or how information is reported and available to the business groups.

Business analysts working in IT carefully consider the context for any information technology change. They consider whether the change is managed as a project, a continuous improvement, or a maintenance activity. Business analysts also consider organizational change management and all impacts including training, communications, and adoption of the change.

The nature of business analysis activities in an IT environment depend on a variety of solution impact factors:

- What happens to the business if this system shuts down?
- What happens if the system performance degrades?
- What business capabilities and processes depend on the IT system?
- Who contributes to those capabilities and processes?
- Who uses those capabilities and processes?

When considering these solution impact factors, not only do business analysts match the formality of analysis activities to the business analysis processes defined by the organization, but also consider the importance of the IT system. The importance of the system under analysis may indicate that more analysis is needed to support and define the requirements for the change.

.2 Depth of Change

Changes in an IT environment frequently require the business analyst to define explicit details, including technical details such as the definition of individual data elements being manipulated or impacted by the change. Integration efforts can require analysis and definition at a great level of detail while identifying and defining the interfaces between IT systems. Due to the level of detail required in these types of initiatives, business analysts elicit and analyze how the organization works as a whole and how the IT system will support those operations. This provides the necessary context for the business analyst to understand whether the details being discovered and documented are relevant to delivering value. This can be particularly challenging when an IT system change is initiated for technology driven reasons but without sufficient clarity or alignment to business purpose.

.3 Value and Solutions Delivered

Information technology systems are implemented to increase organizational value, which includes any support capabilities and processes that use the system. Business analysts seek to align IT functionality to these processes and capabilities, and to measure the effect that the system has on them.

Changes to IT systems can increase value many ways, including:

- reducing operating costs,
- decreasing wasted effort,

- increasing strategic alignment,
- increasing reliability and stability,
- automating error-prone or manual processes,
- repairing problems,
- making it possible to scale up, enhance, or make more readily available a business capability, and
- implementing new functionality and new capabilities.

.4 Delivery Approach

The delivery of business analysis activities within an IT organization varies greatly. Initiatives may range from small enhancement efforts which are completed with a single, short time frame release schedule to multi-release, phased implementations.

Short time frame initiatives may involve a single business analyst for a short period of time. Larger efforts frequently involve several business analysts who may coordinate analysis activities in several ways. Business analysts may divide work based on business group involved or by specific activity.

.5 Major Assumptions

The following is a list of major assumptions of the IT discipline:

- business capabilities and processes that use an IT system are delivering value to the organization,
- business analysts working from other perspectives can integrate their work with the work of the IT business analysts, and
- IT systems changes are usually driven by a business need, although some initiatives may originate from within technology developments.

11.3.2

Business Analysis Scope

.1 Change Sponsor

Information technology changes may be requested or sponsored by business sponsors, IT departments, or as a collaboration between the two. These changes should align to organizational strategy and business goals. It is possible for an IT department to initiate change to align with technical strategy or reach technical goals, but an overall organizational strategy alignment is still crucial for change success.

The following list represents possible change sponsors:

- technical team,
- technical executive,
- application owner,

- process owner,
- business owner,
- internal product manager, and
- regulatory representative (such as a corporate legal department).

Enterprises may use many methods to initiate changes related to information technology. Frequently, large enterprises define a program or project management office within the IT department, which intakes requests and prioritizes efforts on behalf of the department.

.2 Change Targets

Business analysts identify all possible departments, processes, applications, and functions which can be impacted by the proposed change. A business analyst not only focuses on details of the initiative, but also keeps an eye on the larger picture and the potential impact (both business and technical) of the change. This involves a level of process and functional analysis with specific focus on both technical interfaces as well as process hand-offs.

.3 Business Analyst Position

Within an IT initiative, the business analysis activities may be filled by personnel with one of several types of backgrounds or job titles within the organization. This assignment may be dependent upon the type of change, the level of experience, knowledge needed, or simply the personnel available to staff the effort. The personnel may be assigned to the business analysis tasks due to the experience described below, and may complete some or all of the business analysis responsibilities for a given change.

It is possible that all business analysis tasks for an IT project may be completed by a person with only one of these backgrounds:

- a business analyst who works specifically with the business users of an IT system,
- an IT business analyst who is the designated liaison between the technical team and the business group which uses the application,
- a subject matter expert (SME) experienced with the current software implementation,
- a software user experienced with the daily activity of how the software is used and can focus on usability,
- a systems analyst who has experience within the business domain, but does not have experience with the specific application,
- a business process owner who has a depth of experience with the business capabilities or processes, but may not have any technical or IT experience,
- a technical person with a depth of technical experience, or

- a COTS representative who will allow for customized implementations of a packaged solution, and leverage the knowledge of the vendor's package and past implementation experience.

.4 Business Analysis Outcomes

Within an IT initiative, a business analyst may consider business processes impacted by the change, as well as the data and business intelligence information collected by the system. Business analysts working in the initiative thoroughly plan the business analysis effort and the deliverables that support the change effort.

The change approach being utilized has a direct impact on business analysis deliverables or outcomes. Many organizations have a defined system or solution development methodology which, to some extent, dictates the deliverables which are required at each project milestone. Even within the context of this structure the business analyst may seek to complete additional deliverables beyond those required by the change approach or organization specific process, and employ techniques which support the comprehensive understanding of the change effort needed.

Business analysts working in the IT discipline are responsible for delivering any of the following:

- defined, complete, testable, prioritized, and verified requirements,
- analysis of alternatives,
- business rules,
- gap analysis,
- functional decomposition,
- use cases and scenarios, and/or user stories as appropriate,
- interface analysis,
- prototypes,
- process analysis,
- process models,
- state models,
- decision models,
- context models or scope models, and
- data models.

Additional deliverables not included in the above list but relating to any of the outputs of business analysis techniques used may also be considered deliverables of the business analyst.

11.3.3 Methodologies

The methodologies followed by information technology organizations vary widely.

In general, solution development methodologies fall into two generic approaches:

- **Predictive:** structured processes which emphasize planning and formal documentation of the processes used to complete the change. Each phase of the process or sequence is completed before advancing to the next phase.
- **Adaptive:** processes which allow for reworking within one or more of the overall structured process cycles. Most adaptive models are both iterative and incremental, focusing on growing the product in both breadth and depth.

A hybrid methodology may also be utilized. A hybrid may include an overall vision for the whole initiative (as in predictive), as well as a definition of details within individual cycles or iterations (as in adaptive).

The following table identifies several established methodologies or approaches that a business analyst practicing in an information technology environment may encounter.

Table 11.3.1: Information Technology Methodologies

Methodology	Brief Description
Homegrown or Organization Specific	A methodology which is derived from components of other established methodologies or approaches may be created by an information technology organization to govern information technology based initiatives.
Requirements Engineering (RE)	Establishes a structured approach for requirements development and management and is used in predictive, adaptive, and agile environments.
Structured Systems Analysis and Design Method (SSADM)	A predictive development methodology that focuses on established logical modelling and the separation of requirements from solutions as central to systems analysis and specification.
Unified Process (UP)	An adaptive development approach. The inception and elaboration phases are of particular interest to business analysts. UP is not considered agile but is an adaptive methodology.

11.3.4 Underlying Competencies

A business analyst working within IT may possess skills related to IT development such as programming, creating a database, creating a system or solution

architecture, software testing experience, or other technical skills. However, development-related skills or technical skills are not necessary for a business analyst to be successful within an IT environment. It is important for the business analyst to have a strong understanding of the detail required within a requirements package to support technical solutions, as well as an understanding of what is technically feasible within the constraints of an organization's technical architecture. These skills will enable a business analyst to work with all stakeholders to design a business solution framework which will also allow the technical team the flexibility to design a technical solution.

Business analysts use influencing and facilitation skills when working with stakeholders. Negotiation skills are frequently used when working with business and technical staff to come to agreements and decisions if the costs of a solution (either in budget, time, or architectural impact) conflict with the desired business outcome.

Systems thinking is a crucial competency for business analysts practicing in an IT environment. Systems thinking supports the ability of the business analyst to see the larger picture including any other applications or technical aspects which may be impacted, the details of the specific need, and possible technical solutions. Systems thinking also supports the ability to identify impacts to people, processes, and software which are not necessarily directly changed as part of an IT development effort, and to analyze the risks and possible outcomes of those impacts.

11.3.5 Impact on Knowledge Areas

This section explains how specific business analysis practices within information technology are mapped to business analysis tasks and practices as defined by the *BABOK® Guide*. It also describes how each knowledge area is applied or modified within the IT discipline.

Each knowledge area lists techniques relevant to an IT perspective. Techniques used in the discipline of information technology do not deviate, to any great extent, from the *BABOK® Guide* techniques. *BABOK® Guide* techniques are found in the Techniques chapter of the *BABOK® Guide*. This is not intended to be an exhaustive list of techniques but rather to highlight the types of techniques used by business analysts while performing the tasks within the knowledge area.

.1 Business Analysis Planning and Monitoring

A business analysis approach is a fundamental communication tool which can be used to identify resources required for business analysis work and ensure adequate time for the analysis effort. A well-defined business analysis plan integrates into the overall project plan and provides business analysts with the opportunity to define and schedule the business analysis activities for the project.

Many organizations have some standards and processes in place, which may identify certain analysis tasks and deliverables. If these are not in place, the business analyst identifies these tasks and deliverables based on the needs of the specific initiative.

It is important that the context of the analysis work is understood. This includes understanding the inter-operation of software systems, business processes, and the data that is passed from one system to the next. Changes to any single system or process may have a ripple effect that brings additional systems, processes, or stakeholder groups into the scope of the initiative.

The IT business analyst may be embedded within a software team. This approach allows the business analyst to become quite knowledgeable about specific software or processes supported by the software. Stakeholder attitudes and needs may change or shift in regards to each particular change. Roles, collaboration, and communication plans are planned for every change effort.

COTS solutions can involve major systems integration efforts, customizations, and many unexpected tasks due to the introduction of external software. When planning for unknown impacts and unknown customization needs, business analysts engage both internal stakeholders who understand the needs of the change, and external stakeholders who have expertise with the COTS solution being implemented.

BABOK® Guide Techniques

- Backlog Management (p. 220)
- Document Analysis (p. 269)
- Estimation (p. 271)
- Functional Decomposition (p. 283)
- Item Tracking (p. 294)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Organizational Modelling (p. 308)
- Roles and Permissions Matrix (p. 333)
- Scope Modelling (p. 338)
- Stakeholder List, Map, or Personas (p. 344)

.2 Elicitation and Collaboration

Information technology changes frequently affect many stakeholders with distinct relationships to the solution or change. When a change involves an IT application or system, the technical staff may have expertise, perspectives, or experience that can identify additional impacts to systems or processes as requirements and solutions are defined. For this reason, it is beneficial to have at least one elicitation session with IT technical personnel, such as development or technical design staff, and business SMEs in the same room at the same time. This type of elicitation approach provides a platform for collaboration between technical and business teams, where the IT business analyst serves as a facilitator and liaison for the process.

Business analysts practicing in an IT environment may utilize any of the techniques identified in the Elicitation and Collaboration knowledge area. Additionally, the following methods can be of great benefit in the information technology discipline:

- **Investigation:** using organizational process assets, market research, competitive analysis, functional specifications, and observation,

- **Simulations:** using statistical modelling and mock-ups, and
- **Experimentation:** using proofs of concept, prototypes, alpha- and beta-releases, and A/B testing.

Information technology changes can be seen as a distraction or cost by business stakeholders if the change is not perceived as mission critical or if the stakeholder is experiencing negative value from the change. This can make engagement for elicitation challenging. Elicitation across organizational boundaries may be impeded, causing collaboration breakdowns and rework. IT business analysts can decrease the risk of rework by engaging information technology and business resources in collaboration activities.

BABOK® Guide Techniques

- Brainstorming (p. 227)
- Collaborative Games (p. 243)
- Document Analysis (p. 269)
- Focus Groups (p. 279)
- Interface Analysis (p. 287)
- Interviews (p. 290)
- Observation (p. 305)
- Process Modelling (p. 318)
- Prototyping (p. 323)
- Scope Modelling (p. 338)
- Sequence Diagrams (p. 341)
- Stakeholder List, Map, or Personas (p. 344)
- State Modelling (p. 348)
- Survey or Questionnaire (p. 350)
- Use Cases and Scenarios (p. 356)
- Workshops (p. 363)

.3 Requirements Life Cycle Management

IT initiatives frequently experience major discoveries while creating the change. It is through exploration that the business analysts discover the implications of the new functionality provided by the solution. This sense of discovery in IT environments has led to the adaptation of short cycle times (agile and continuous improvement), rigorous change control (Capability Maturity Model Integration (CMMI) and predictive), and externalized information technology (Software as a Service (SaaS) and cloud services).

Business analysts working in IT pay particular attention to alignment, approval, change control, traceability, and requirements life cycle management tools. It is the role of the business analyst to work with stakeholders to develop a consistent method for reviewing evolving requirements to ensure alignment with the business objectives for the initiative.

In many cases, changes to approved requirements are driven by changes to higher-level requirements such as business objectives. Business analysts collaborate with stakeholders to ensure these requirements are stable before proceeding to solution or technical requirements. When changes to requirements are presented, the business analyst analyzes the impact and plans how to manage proposed changes.

As the complexity of an information technology environment grows, it becomes increasingly important to track each change to each requirement or between

requirements and other information. Traceability that includes dependencies and relationships among requirements makes it easier for stakeholders to understand what is changing about the IT system and predict impacts of additional changes.

As technical systems are changed over time, it is helpful when each version of each requirement is stored in some way and accounted for. Traceability makes it possible to find the source and owner of each requested function and feature, as well as why, when, and how it changed over time. This history is important for ensuring that the requirements are complete and that the approval of requirements is a sensible decision. When the change-work and the IT system are audited, regulators and other interested parties can understand what happened, when, and why. This can be especially important for audit purposes, when an application manages data or processes systematically without human intervention for each transaction or instance of the process occurring. This tracing also helps the organization understand why some functionality is not delivered or implemented in the IT system, and why it was dropped from the scope of this implementation.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Decision Analysis (p. 261)
- Prioritization (p. 311)
- Item Tracking (p. 294)

.4 Strategy Analysis

Within an IT organization, strategy analysis focuses on the technologies and systems, business units, business processes, and business strategies impacted by a proposed change. It is possible that the impacts of a change cause a ripple effect through other systems in the organization. In order to analyze needs and proposed changes, business analysts seek to understand all the various aspects that may be impacted by the change.

Current state analysis within IT initiatives includes analysis of manual processes, understanding what the system or technology currently does, the data needed to complete tasks, and the other systems and processes that interact with the system. Business analysts plan for a thorough understanding of the current state and a large context of the enterprise at first, with the understanding that the scope will narrow as the future state is identified.

Once the current state is understood, the desired future state is described. This may be process or capability related and usually includes how current system functionality is required to change in order to support the future vision and meet the objectives of both individual stakeholders and the enterprise. In understanding both the current and future states, the gap between the two is identified, and that is where the direction of the change effort can be set. It is at this point of analysis that solution options are explored.

Once the aspects of the change scope and desired future state are understood, business analysts assess uncertainty and risk. Uncertainty is clarified by:

- identifying and defining risks,
- identifying and defining potential benefits,
- establishing parameters for variance in known processes and operations, and
- exploring the unknown.

Business analysts also explore other potential risks including:

- vendor risks, such as their business and product stability,
- impacts to the system's technical environment,
- scalability of the solution should volumes of transactions or users increase over time, and
- additional process or system changes required based on the change initiated.

BABOK® Guide Techniques

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|---|------------------------------------|
| • Business Capability Analysis (p. 230) | • Process Modelling (p. 318) |
| • Focus Groups (p. 279) | • Scope Modelling (p. 338) |
| • Functional Decomposition (p. 283) | • Survey or Questionnaire (p. 350) |
| • Interviews (p. 290) | • SWOT Analysis (p. 353) |
| • Item Tracking (p. 294) | • Vendor Assessment (p. 361) |
| • Observation (p. 305) | • Workshops (p. 363) |
| • Process Analysis (p. 314) | |

.5 Requirements Analysis and Design Definition

It is important for business analysts working in IT to understand and clarify the term 'design'. Many IT organizations think of design only as it applies to the design or blueprint of a software or technical change. Within the Requirements Analysis and Design Definition knowledge area, the term design is viewed more broadly and from the business analyst's point of view. Designs are usable representations that focus on the solution and understanding how value might be realized by a solution if it is built. For example, a model of a potential process improvement (whether it impacts or utilizes an IT system or not), as well as user interface layouts or report definitions, can all be considered designs.

Business analysts elaborate business and technical requirements, break down and define stakeholder needs, and identify the value to be realized by stakeholders once a technical solution or change is implemented. They elicit, define, and analyze business and stakeholder requirements, and also define, analyze, and model solution designs. They define requirements to a level of technical detail that will be used as part of solution design and input into technical designs. This elaboration will include both functional requirements and non-functional

requirements. For some change initiatives, the definition of non-functional requirements could define all business goals for the change effort.

Business analysts often rely on other change agents to produce technical designs for software solutions. A systems architect, programmer, database manager, or other technical expert is often needed to determine how to use technology to satisfy a set of requirements. IT business analysts define process steps, business rules, screen flows, and report layouts. Defining requirements to include detailed functionality of a system, the business, and system processes is a crucial part of solution design and does not separate analysis and design.

As part of requirements analysis, an IT business analyst may partner with another business analyst with a different focus, such as an enterprise business analyst or business architect, to ensure that the IT requirements align to business or organizational strategy.

Requirements analysis and design definition frequently involves documenting requirements using words and pictures. In some cases, requirements may be represented in other ways such as a proof of concept, working software prototypes, or simulations. In all cases, the business analyst works to produce documentation with sufficient and appropriate details for:

- the business to verify and validate the requirements,
- the developers to design from, and
- the testers to measure the solution against before it is implemented into a production environment.

BABOK® Guide Techniques

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| • Business Rules Analysis (p. 240) | • Non-Functional Requirements Analysis (p. 302) |
| • Data Dictionary (p. 247) | • Organizational Modelling (p. 308) |
| • Data Flow Diagrams (p. 250) | • Process Modelling (p. 318) |
| • Data Modelling (p. 256) | • Prototyping (p. 323) |
| • Decision Analysis (p. 261) | • Reviews (p. 326) |
| • Decision Modelling (p. 265) | • Roles and Permissions Matrix (p. 333) |
| • Document Analysis (p. 269) | • Scope Modelling (p. 338) |
| • Estimation (p. 271) | • Sequence Diagrams (p. 341) |
| • Functional Decomposition (p. 283) | • State Modelling (p. 348) |
| • Glossary (p. 286) | • Use Cases and Scenarios (p. 356) |
| • Interface Analysis (p. 287) | • User Stories (p. 359) |

.6 Solution Evaluation

Solution evaluation focuses on solution components and the value they provide. Within an IT context, this includes a focus on the interactions between multiple

systems within the change and the surrounding environment. It is important for a business analyst working in the IT discipline to understand the context of the solution and how changes within one system or process can impact other systems within the environment. These impacts can add negative or positive value to the other systems, therefore impacting the overall realization of value for the change.

One aspect of solution evaluation within an IT context is software testing or solution testing. Testing or quality assurance ensures that the solution performs as anticipated or designed, and that it meets the needs of the business or stakeholders who requested the change effort. The business analyst works with quality assurance (testers) to ensure that technical solutions will meet the business needs as defined by the requirements and other business analysis deliverables. Testers utilize testing methodologies to plan, develop, and execute tests. This aspect of solution testing generally focuses on complete process testing, including across systems to ensure end-to-end solution quality and accuracy. Business analysts work with stakeholders to plan, develop, and execute user acceptance tests to ensure that the solution meets their needs.

Business analysts make themselves aware of the rationale for implementing an IT solution and how that rationale works to create solution value. This value realization is commonly associated with better support for business processes and procedures.

Business and technical objectives are associated with benefits and value realization which are measured against defined metrics used to evaluate success. Requirements should trace back to the objectives, and this traceability provides a foundation for solution evaluation. The analysis of solution performance focuses on technical systems and how they provide potential and actual value to stakeholders.

Where a large organizational change contains an IT element, an IT solution evaluation can contribute to a broader benefits realization activity associated with the whole change program.

As part of solution evaluation activities, a business analyst may work with a team to complete tasks, such as assessing solution limitations and assessing the impacts of such limitations. The business analyst may support and assess technical testing efforts for all, or a portion of, the developed solution.

BABOK® Guide Techniques

- Acceptance and Evaluation Criteria (p. 217)
- Decision Analysis (p. 261)
- Estimation (p. 271)
- Item Tracking (p. 294)
- Metrics and Key Performance Indicators (KPIs) (p. 297)
- Organizational Modelling (p. 308)
- Risk Analysis and Management (p. 329)
- Process Modelling (p. 318)
- SWOT Analysis (p. 353)
- Vendor Assessment (p. 361)